## Investigation: METALURGICA DUQUE SA (CVM code 8648), fiscal year 2011, as it was on 2013-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

## Company

Sector: Metalurgia e Siderurgia (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

## Document used

2011 DFP, version 1, filed on 2012-03-30; origin: open data; type: individual; original scale: MIL.
Versions known on the date: 1 (2012-03-30).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=15754

## Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −1.866 (index)
- Alert by the study 1 rule: no
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: 91 (n = 17)
- Percentile in group and year: 82 (n = 363)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 1.908 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 93 (n = 23)
- Percentile in group and year: 93 (n = 457)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 0.862 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 2 (n = 23)
- Percentile in group and year: 17 (n = 467)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 0.816 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 20 (n = 23)
- Percentile in group and year: 12 (n = 462)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 1.065 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 59 (n = 23)

Percentile in group and year: 38 (n = 468)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: 1.059 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 62 (n = 17)
Percentile in group and year: 71 (n = 377)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 1.002 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 46 (n = 23)
Percentile in group and year: 47 (n = 461)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.013 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 54 (n = 23)
Percentile in group and year: 63 (n = 484)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 1.242 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 85 (n = 23)
Percentile in group and year: 84 (n = 476)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: 28.0 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 34.2 days)
Percentile in sector and year: 93 (n = 22)
Percentile in group and year: 88 (n = 441)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 11.87 p.p.
Alert by the study 1 rule: yes
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 63 (n = 23)
Percentile in group and year: 67 (n = 429)
The company's own history: fewer than 5 previous fiscal years with a value

Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.014 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 59 (n = 23)
Percentile in group and year: 63 (n = 484)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

## Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: 0.779 (index)
Components: X1 = −0.127; X2 = 0.061; X3 = 0.063; X4 = 0.943 (indices)
Percentile in sector and year: 27 (n = 24)
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

## Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 7/9 (8 computable)
Signals:
- Return on assets (ROA): 1
- Operating cash flow on assets (CFO): 1
- Change in ROA: 1
- Cash above earnings (ACCRUAL): 1
- Change in leverage: 1
- Change in current ratio: 1
- No equity issuance (EQ_OFFER): —
- Change in gross margin: 1
- Change in asset turnover: 0
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

## Changes

Highlighted in the following year's comparative: none.
Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets of the version used. The other differences (reclassifications between lines and smaller differences) stay in the full list.
Full list: 1 accounts above the R$1,000 tolerance (group a, totals and key lines: 0; group b, reclassifications: 1), in the annex at the end.

## What was known on the date

- DFP 2010, v1: 2011-03-22
- FCA 2011, v1: 2011-03-22
- FCA 2011, v2: 2011-06-03

- DFP 2011, v1: 2012-03-30
- DFP 2012, v1: 2013-03-28
DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

## Data quality
53 applicable checks; 0 failed.

## Auditor's report and auditor (factual section; no alert)
Auditor's report: Sem Ressalva (unqualified).
no opinion excerpt was found in the text
Auditor in the 2010 FCA: Audiplan Auditores Independentes.
Auditor in the 2011 FCA: Audiplan Auditores Independentes.
Auditor in the 2012 FCA: Audiplan Auditores Independentes.
Auditor changes: none in the FCAs filed by the date.
Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

## Document excerpts (copied; no alert)
Excerpts copied from the documents, with the page, and the Reference Form table. It is not a signal family: it generates no alert, score or ranking. Absence of mention does not prove absence.
How the reader was tested: 20 held-out cases, a single measurement with the frozen reader (c99c953): precision 0.962, coverage 0.962, literal check on 132 of 132 excerpts, no natural-person name (docs/leitor_medida.json).

Explanatory notes · Related parties
The reader found no note with "partes relacionadas" (related parties) in the title.

Explanatory notes · Reverse factoring and forfaiting
The reader found no mention of reverse factoring (risco sacado), forfaiting or supplier agreements in the notes of this DFP. Absence of mention does not prove absence of the operation.

Auditor's report
Type of opinion: unqualified

Reference Form
The Reference Form for this fiscal year was not available on the date. It is the only version available in the open data; earlier versions are not there.

Limits of this section:
- In real documents, about 22% of the excerpts come out omitted by the name detector, including names of companies outside the registry.
- In tables, header lines (years, column titles) may come out as excerpts; the held-out cases had no tables with multi-line headers.

## How the engine was tested
In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

## Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

## Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 38 DFP lines, each with document, version, filing date, account and location.

## Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a F1); leitor-f1-1 (commit c99c953). Generated at 2026-11-12T03:41:10+00:00.

## Today's information (outside the reference date)

Registry status: CANCELADA. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: METALURGICA DUQUE SA - EM RECUPERAÇÃO JUDICIAL.

## Annex: full list of differences in the following year's comparative (R$ million)

(b) DRE 3.99.01.01 ON: 0.0 in the version used, 0.0 in the comparative (difference of −0.0)